Software Requirements Specification

TradeFlow — Intelligent Logistics & Freight Matching Platform for Landlocked Ethiopia

Prepared for Ethiopian Shipping and Logistics Services Enterprise / Freight Forwarders Association

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1 1. Introduction

1.1 1.1 Purpose

This SRS specifies the requirements for **TradeFlow**, a digital freight marketplace and logistics-optimization platform connecting cargo owners with transporters along Ethiopia's principal trade corridor (Djibouti Port → Modjo Dry Port → regional hubs → final destination). It is written for the development team, the Ethiopian Shipping and Logistics Services Enterprise, freight forwarders, customs authorities, and independent transporters, and follows an IEEE 830-style structure.

1.2 1.2 Scope

TradeFlow will:

- Operate a two-sided marketplace matching shippers/importers/exporters with transporters and freight forwarders.
- Provide real-time shipment tracking and ETA prediction across the full corridor.
- Offer dynamic spot and contract pricing responsive to fuel costs and corridor congestion.
- Optimize routing under fuel-scarcity, road-condition, and security constraints.
- Digitize customs documentation and automate compliance validation.
- Provide settlement via mobile money and reporting/dispute resolution tooling.

Out of scope for v1.0: bonded-warehouse management systems, vessel booking with international shipping lines (assumed to occur upstream), and customs duty calculation engines beyond validation of submitted documentation.

1.3 1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
CVRP	Capacitated Vehicle Routing Problem
ETA	Estimated Time of Arrival
Freight Forwarder	Intermediary arranging transport/customs on behalf of a shipper
Modjo Dry Port	Ethiopia's principal inland customs and freight terminal
Spot Rate	Price for a one-off shipment negotiated at time of booking
Contract Rate	Pre-negotiated recurring price between a shipper and transporter
TMS	Transport Management System

1.4 1.4 References

- IEEE Std 830-1998
- Ethiopian Customs Commission clearance procedure documentation
- Ethiopian Roads Authority road-condition reporting standards

1.5 1.5 Document Overview

Sections follow the same structure as a standard SRS: overall description (2), detailed features (3), interfaces (4), non-functional requirements (5), data model (6), architecture (7), algorithms (8), appendices (9).

2 2. Overall Description

2.1 2.1 Product Perspective

TradeFlow is a new, independent SaaS marketplace. It complements — rather than replaces — the Ethiopian Shipping and Logistics Services Enterprise's operational systems, acting as a coordination, transparency, and optimization layer across public and private logistics actors on the Djibouti–Ethiopia corridor.

2.2 2.2 Product Functions (Summary)

1. Shipper/transporter onboarding, verification, and rating
2. AI-assisted freight matching
3. GPS-based shipment tracking with ETA prediction
4. Dynamic spot/contract pricing
5. Fuel- and security-aware route optimization
6. Digital customs documentation and validation
7. Fuel consumption analytics
8. Conflict-zone alerts and rerouting
9. Transporter performance analytics
10. Mobile-money payment and settlement

2.3 2.3 User Classes and Characteristics

User Class	Description	Technical Proficiency	Primary Device
Shipper/Importer/Exporter	Posts cargo, books transporters, tracks shipments	Medium	Web
Transporter/Owner-Operator	Accepts loads, manages fleet, receives payment	Low–Medium	Mobile App
Driver	Executes trips, checks in at waypoints, reports incidents	Low	Mobile App (offline)
Freight Forwarder	Manages multiple shippers' cargo and documentation	Medium–High	Web
Customs Authority Staff	Reviews and validates clearance documentation	Medium	Web
Platform Administrator	Manages disputes, compliance, and system configuration	High	Admin Console

2.4 2.4 Operating Environment

- Web dashboard: modern browsers, desktop/tablet.
- Driver and transporter apps: Android-first (majority device share), iOS secondary.

- Backend hosted on cloud infrastructure with regional (Ethiopia/East Africa) data residency for compliance and latency.

2.5 2.5 Design and Implementation Constraints

- Must function under intermittent connectivity along highway corridors with sparse network coverage; core check-in and tracking features require offline queuing.
- Must integrate with fragmented, sometimes manual, customs and fuel-availability data sources.
- Payment settlement dependent on mobile-money network availability and reliability.

2.6 2.6 Assumptions and Dependencies

- Ethiopian Customs Commission will make clearance-status data available via API or a secure batch interface.
- Fuel-station availability data will be sourced from a combination of partner APIs and crowd-sourced driver reports.
- GPS hardware (in-app or aftermarket trackers) will be available in the majority of participating fleets.

3 3. System Features and Detailed Functional Requirements

3.1 3.1 FR-01 — Two-Sided Marketplace Onboarding (Critical)

Sub-requirements: - FR-01.1: Shippers and transporters register with business/legal identity verification (trade license number, tax ID). - FR-01.2: Transporters submit fleet details (vehicle type, capacity, fuel type, insurance, roadworthiness certificate) for verification before activation. - FR-01.3: Post-trip rating (1–5 stars plus comments) collected from both parties; average rating displayed on profile. - FR-01.4: Verification status (Pending/Verified/Suspended) visible to platform administrators with an audit trail of status changes.

Acceptance Criteria: An unverified transporter cannot accept bookings; the system blocks the "Accept Load" action and surfaces a clear verification-status message.

3.2 3.2 FR-02 — Real-Time AI-Powered Freight Matching (Critical)

Sub-requirements: - FR-02.1: Shippers post load details (origin, destination, cargo type, weight/volume, required pickup window). - FR-02.2: Matching engine ranks eligible transporters by a weighted score of cost, historical reliability, fuel efficiency, and proximity/availability. - FR-02.3: Shippers may accept a suggested match or open the load to bidding among a shortlist of qualified transporters. - FR-02.4: Matching response time target: recommendations returned within 1 second for a load posting under normal load conditions.

3.3 3.3 FR-03 — Shipment Tracking & ETA Prediction (Critical)

Sub-requirements: - FR-03.1: GPS position updates ingested at least every 5 minutes while a shipment is in transit (subject to connectivity). - FR-03.2: ETA model recalculates predicted arrival using live position, historical corridor transit-time data, weather, and known congestion/conflict alerts. - FR-03.3: Shippers receive milestone notifications (departed origin, cleared customs, arrived destination) via app, SMS, or email per their preference.

3.4 3.4 FR-04 — Dynamic Pricing Engine (High)

Sub-requirements: - FR-04.1: Spot rates computed from current demand/capacity balance, fuel-cost index, and corridor congestion level, refreshed at least hourly. - FR-04.2: Contract-rate shippers may lock a rate for a defined period; system flags contract rates that diverge materially from prevailing spot rates for renegotiation review. - FR-04.3: All price computations are logged with the inputs used, for auditability and dispute resolution.

3.5 3.5 FR-05 — Fuel- and Security-Aware Route Optimization (High)

Sub-requirements: - FR-05.1: Recommend routes minimizing a weighted cost of distance, time, fuel consumption, and risk exposure, given current fuel-station stock and flagged security zones. - FR-05.2: Support driver-reported, geotagged incident logging (accident, checkpoint delay, fuel unavailability) that updates the routing model in near-real time. - FR-05.3: Automatically propose rerouting when a materially better path becomes available mid-trip; require driver/dispatcher confirmation before altering an active route.

3.6 3.6 FR-06 — Digital Customs Documentation (High)

Sub-requirements: - FR-06.1: Support upload and structured capture of key clearance documents (commercial invoice, packing list, bill of lading, certificate of origin). - FR-06.2: Automated validation checks for completeness and consistency (e.g., matching quantities/values across documents) before submission. - FR-06.3: Status tracking of clearance (Submitted/Under Review/Cleared/Rejected) with rejection reasons surfaced to the shipper/forwarder.

3.7 3.7 FR-07 — Fuel Consumption Analytics (High)

Sub-requirements: - FR-07.1: Track estimated vs. actual fuel consumption per trip (where telematics data is available) and surface efficiency trends per vehicle/driver. - FR-07.2: Recommend driving/route adjustments to reduce fuel consumption based on historical patterns.

3.8 3.8 FR-08 — Conflict Zone Alerts & Rerouting (Medium)

Sub-requirements: - FR-08.1: Maintain a continuously updated set of flagged risk zones sourced from government advisories and verified crowdsourced reports. - FR-08.2: Push proactive alerts to drivers and dispatchers approaching a flagged zone, with suggested alternate routing.

3.9 3.9 FR-09 — Transporter Performance Analytics (Medium)

Sub-requirements: - FR-09.1: Track on-time delivery rate, incident rate, fuel efficiency, and customer rating per transporter, aggregated monthly. - FR-09.2: Provide transporters a self-service dashboard benchmarking their performance against anonymized corridor averages.

3.10 3.10 FR-10 — Payment and Settlement (Medium)

Sub-requirements: - FR-10.1: Support mobile-money-based payment initiation and confirmation (TeleBirr and equivalents) with automated reconciliation against completed trips. - FR-10.2: Support platform commission calculation and automated payout scheduling to transporters. - FR-10.3: Provide a dispute-flagging workflow for payment discrepancies, routed to platform administrators.

4 4. External Interface Requirements

4.1 4.1 User Interfaces

- **Web Dashboard:** For shippers, freight forwarders, and administrators — load posting, tracking map, document upload, analytics.
- **Transporter Mobile App:** Load browsing/bidding, fleet management, earnings dashboard.
- **Driver Mobile App:** Turn-by-turn navigation with offline caching, waypoint check-in, incident reporting, digital proof-of-delivery capture (signature/photo).
- **Admin Console:** Verification queue, dispute resolution, compliance monitoring, system health.

4.2 4.2 Hardware Interfaces

- Smartphone GPS (primary); optional integration with aftermarket fleet-tracking hardware via standard telematics protocols (e.g., OBD-II/GPS trackers with REST/MQTT output).

4.3 4.3 Software Interfaces

Interface	Protocol	Direction	Data
Ethiopian Customs Commission	REST/secure file transfer	Bidirectional	Clearance status/documents
Ethiopian Roads Authority	REST/batch	Inbound	Road condition data
Fleet GPS/telematics providers	REST/MQTT	Inbound	Position, fuel telemetry
Fuel station partner network	REST	Inbound	Fuel availability
TeleBirr and mobile-money providers	REST/Webhook	Bidirectional	Payments

4.4 4.4 Communications Interfaces

- TLS 1.2+ for all API traffic; MQTT over TLS for high-frequency telematics streams.
- Push notifications via FCM/APNs for mobile apps; SMS fallback for low-connectivity users.

5 5. Non-Functional Requirements

5.1 5.1 Performance

- Matching algorithm response: < 1 second (P95) for a standard load posting.
- Tracking update latency: < 5 seconds from device transmission to dashboard display under normal connectivity.

5.2 5.2 Availability & Reliability

- 99.9% uptime SLA for tracking and matching services (core revenue-generating functions).
- Offline queuing for driver check-ins and incident reports, syncing automatically upon reconnect with conflict resolution favoring most-recent authoritative timestamp.

5.3 5.3 Scalability

- Support 50,000+ active shipments per month and 10,000+ concurrent users at peak (e.g., end-of-month clearance surges).
- Auto-scaling for matching and tracking microservices based on load-posting and GPS-ingestion throughput.

5.4 5.4 Security

- Role-based access control; encrypted communications (TLS in transit, AES-256 at rest).
- Fraud-detection rules for suspicious bidding patterns, duplicate accounts, or falsified documentation, with manual review escalation.
- Full audit logging of pricing decisions, document submissions, and payment transactions.

5.5 5.5 Compliance

- Alignment with Ethiopian Customs Commission data-handling requirements.
- Data residency within Ethiopia or an approved regional cloud zone.

5.6 5.6 Usability

- Driver app optimized for one-handed, glance-based use while stationary; minimal text entry, icon-driven navigation.
- Multi-language support (Amharic, English at minimum; Afaan Oromo recommended).

6 6. Data Requirements (High-Level Data Model)

Core Entities: `Shipper`, `Transporter`, `Vehicle`, `Driver`, `Load`, `Bid`, `Shipment`, `TrackingEvent`, `Route`, `PriceQuote`, `CustomsDocument`, `Payment`, `Rating`, `IncidentReport`.

Key relationships: - A `Load` receives many `Bid` records from `Transporter` entities; an accepted `Bid` produces a `Shipment`. - A `Shipment` is associated with a `Route`, a sequence of `TrackingEvent` records, one or more `CustomsDocument` records, and a `Payment`. - `Rating` entities are bidirectional (shipper→transporter and transporter→shipper), linked to a completed `Shipment`.

Data retention: shipment, document, and payment records retained a minimum of 7 years for tax/audit purposes; raw GPS tracking retained 1 year, then aggregated into route-performance statistics.

7 7. System Architecture Overview

- **Presentation Layer:** Web app (shippers/forwarders/admin), transporter app, driver app.
- **API Layer:** REST/GraphQL gateway, RBAC middleware, rate limiting.
- **Application Services:** Onboarding/Verification Service, Matching Service, Tracking/ETA Service, Pricing Service, Routing Service, Customs Document Service, Payment Service, Notification Service, Ratings/Performance Service.
- **Data Layer:** Relational database (transactional), time-series store (GPS/telemetry), object storage (documents/photos), search index (load/transporter discovery).
- **ML/Analytics Layer:** ETA prediction models, pricing models, fraud-detection models, batch analytics jobs.
- **Integration Layer:** Adapters for customs, roads authority, fuel network, telematics providers, mobile money.

8 8. Core Algorithm Specifications

8.1 8.1 Freight Matching

Multi-objective scoring function combining normalized cost, historical on-time-delivery reliability, fuel-efficiency rating, and current proximity/availability, with configurable weights per shipper preference (e.g., cost-optimized vs. speed-optimized). Ranked shortlist returned; shipper may accept top match or open shortlist to competitive bidding.

8.2 8.2 ETA Prediction

Gradient-boosted regression (or comparable) model trained on historical corridor transit times, segmented by route leg, vehicle type, time of day/week, weather, and known congestion/conflict flags; live GPS position used to recompute remaining-leg ETA incrementally as the trip progresses.

8.3 8.3 Dynamic Pricing

Spot price computed as a base corridor rate adjusted by real-time demand/capacity ratio, a fuel-cost index multiplier, and a congestion surcharge/discount; bounded by configurable floor/ceiling values to prevent extreme volatility, with all adjustments logged for transparency and dispute resolution.

8.4 8.4 Route Optimization

Time-dependent capacitated vehicle routing formulation where edge costs vary with fuel availability, road condition, and security-risk flags; solved with a metaheuristic search (e.g., large neighborhood search) for near-optimal routes within seconds, with incremental re-optimization triggered by significant constraint changes during an active trip.

9 9. Appendices

9.1 9.1 Priority Definitions

- **Critical:** Required for MVP launch.
- **High:** Required within first release cycle post-MVP.
- **Medium:** Planned for subsequent releases.

9.2 9.2 Open Issues

- Confirmation needed on the format and update frequency of Ethiopian Customs Commission clearance-status feeds.
- Coverage gaps in GPS/cellular connectivity along parts of the corridor require a defined offline-tolerance policy for ETA accuracy guarantees.

9.3 9.3 Traceability Note

Each functional requirement should be traced to specific test cases and backlog user stories before sprint planning begins.